FREELANCE MARKETPLACE

Entity Relationship (ER) Diagram

Database Design Documentation

Technology Stack: Python + Django + MySQL

Version 1.0 | January 2025

Academic Project Documentation

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1. Introduction

This document presents the Entity Relationship (ER) Diagram for the Freelance Marketplace application. The ER diagram illustrates the database structure, showing all entities (tables), their attributes, and the relationships between them.

The database is designed using MySQL 8.0 and follows Django ORM conventions with UUID primary keys and timestamp fields for audit purposes.

2. Entity Overview

The system consists of 16 core entities organized into functional modules:

Module	Entity	Description
User Management	User	Core user accounts
	FreelancerProfile	Extended freelancer data
	ClientProfile	Extended client data
	Skill	Skill tags
Project Management	Category	Project categories
	Project	Project listings
	ProjectAttachment	Project files
Bidding System	Proposal	Freelancer bids
	ProposalAttachment	Proposal samples
Contract Management	Contract	Active contracts
	Milestone	Contract milestones
	Deliverable	Work deliverables
Messaging	Conversation	Chat threads
	Message	Chat messages
Payments	Transaction	Payment records
	EscrowAccount	Escrow holdings
	FreelancerWallet	Earnings wallet
Reviews	Review	User reviews
Notifications	Notification	User notifications

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3. ER Diagram (Textual Representation)

The following shows the ER diagram in Chen notation style with entities, attributes, and relationships:

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                          FREELANCE MARKETPLACE ER DIAGRAM                    │
└──────────────────────────────────────────────────────────────────────────────┘

                                  ┌──────────┐
                                  │   USER   │
                                  └──────┬───┴──────────────────────┐
                                         │                          │
                          1:1 ▼          1:1 ▼          1:1 ▼
                ┌──────────────────┐ ┌──────────────┐ ┌──────────┐
                │ FREELANCER_PROFILE│ │ CLIENT_PROFILE│ │  WALLET  │
                └──────────────────┘ └──────────────┘ └──────────┘
                         │                  │
                    1:N ▼              1:N ▼
                ┌──────────┐  ◄──►  ┌──────────┐
                │ PROPOSAL │  N:1   │ PROJECT  │
                └──────────┘        └──────┬───┴──────┐
                     │                     │ 1:N      ▼
                1:1 ▼                      ▼
                ┌──────────┐        ┌──────────┐
                │ CONTRACT │        │ATTACHMENT│
                └──────┬───┴────────┬───┴──────┬───┴──────┐
                1:N ▼      1:1 ▼      1:N ▼      1:N ▼
            ┌──────────┐ ┌──────┐ ┌──────────────┐ ┌──────┐
            │ MILESTONE│ │ESCROW│ │CONVERSATION  │ │REVIEW│
            └──────────┘ └──────┘ └──────────────┘ └──────┘
                1:N ▼                  1:N ▼
            ┌──────────┐            ┌──────────┐
            │DELIVERABLE│           │ MESSAGE  │
            └──────────┘            └──────────┘
```

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4. Entity Descriptions

4.1 User Entity

Attribute	Data Type	Constraints	Description
id	UUID	PK	Primary key
email	VARCHAR(254)	UNIQUE, NOT NULL	User email address
username	VARCHAR(150)	UNIQUE, NOT NULL	Username for display
password	VARCHAR(128)	NOT NULL	Hashed password
role	ENUM	NOT NULL	freelancer/client/admin
is_verified	BOOLEAN	DEFAULT FALSE	Email verification status
created_at	DATETIME	AUTO	Account creation date

4.2 Project Entity

Attribute	Data Type	Constraints	Description
id	UUID	PK	Primary key
client_id	UUID	FK → User	Project owner
category_id	UUID	FK → Category	Project category
title	VARCHAR(200)	NOT NULL	Project title
description	TEXT	NOT NULL	Detailed description
budget_type	ENUM	NOT NULL	fixed/hourly/milestone
budget_min	DECIMAL(10,2)	NOT NULL	Minimum budget
budget_max	DECIMAL(10,2)	NOT NULL	Maximum budget
status	ENUM	NOT NULL	Project status

4.3 Contract Entity

Attribute	Data Type	Constraints	Description
id	UUID	PK	Primary key

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project_id	UUID	FK → Project, UNIQUE	Associated project
proposal_id	UUID	FK → Proposal, UNIQUE	Accepted proposal
client_id	UUID	FK → User	Client user
freelancer_id	UUID	FK → User	Freelancer user
total_amount	DECIMAL(10,2)	NOT NULL	Contract value
status	ENUM	NOT NULL	Contract status

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5. Relationship Details

Parent Entity	Child Entity	Cardinality	Description
User	FreelancerProfile	1:1	Each user has one freelancer profile
User	ClientProfile	1:1	Each user has one client profile
User	FreelancerWallet	1:1	Each freelancer has one wallet
User (Client)	Project	1:N	Client can post many projects
User (Freelancer)	Proposal	1:N	Freelancer can submit many proposals
Project	Proposal	1:N	Project receives many proposals
Project	Contract	1:1	Project has one contract
Proposal	Contract	1:1	Accepted proposal creates contract
Contract	Milestone	1:N	Contract has many milestones
Contract	EscrowAccount	1:1	Contract has one escrow account
Contract	Review	1:N	Contract has two reviews (mutual)
Milestone	Deliverable	1:N	Milestone has many deliverables
Milestone	Transaction	1:N	Milestone has payment transactions
Conversation	Message	1:N	Conversation has many messages
User	Notification	1:N	User receives many notifications

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6. Cardinality Summary

6.1 One-to-One (1:1) Relationships

- User ←→ FreelancerProfile
- User ←→ ClientProfile
- User ←→ FreelancerWallet
- Project ←→ Contract
- Proposal ←→ Contract
- Contract ←→ EscrowAccount

6.2 One-to-Many (1:N) Relationships

- User (Client) → Projects
- User (Freelancer) → Proposals
- Project → Proposals
- Project → ProjectAttachments
- Contract → Milestones
- Contract → Reviews
- Contract → Conversations
- Milestone → Deliverables
- Milestone → Transactions
- Conversation → Messages
- User → Notifications
- User → Transactions

6.3 Many-to-Many (M:N) Relationships

- User ←→ Conversation (via participants junction)

— End of Document —

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